

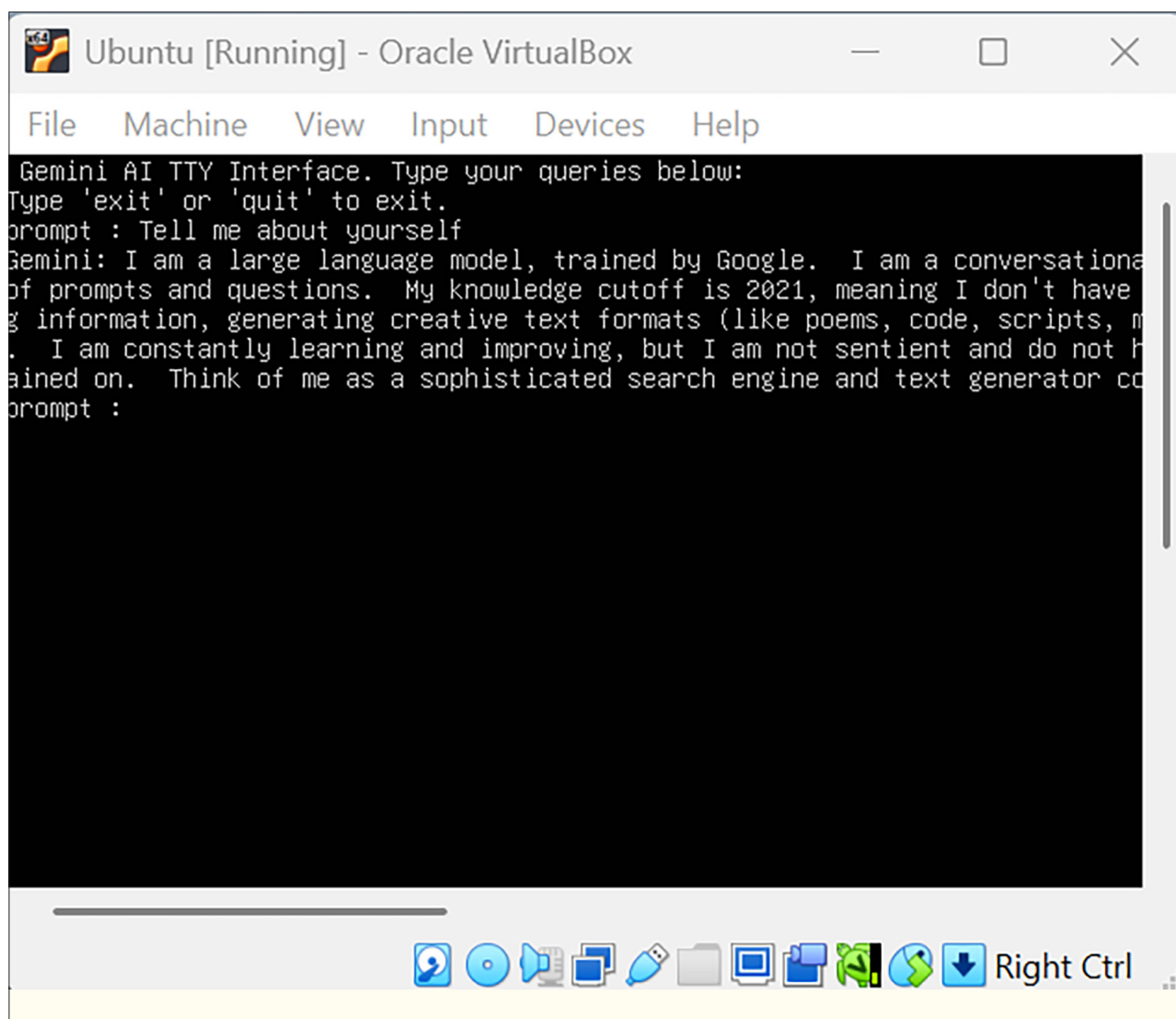


Figure 5: Netbooting a custom LLM-based OS

application will automatically generate a modified disk image for network booting.

Right-click on the *NEW\_PXE* folder, open its properties, and enable network sharing for all users to ensure accessibility.

In VirtualBox, select the *Netboot* option to be top priority in boot order and set other settings as shown in Figure 4.

Soon after booting into the newly created custom OS, an interactive terminal is displayed, enabling seamless AI-based queries. This terminal serves as the interface for users to interact with the AI system, allowing for a wide range of applications. From querying the Gemini API to utilising various networking tools, the terminal facilitates dynamic and real-time communication with the system. The extensive functionality provided by this AI-powered

terminal can be leveraged for tasks such as data analysis, automation, system monitoring, and other complex operations, all of which are integral to the success of the custom OS. As shown in Figure 5, the user is presented with a streamlined, efficient environment for executing advanced queries and commands, demonstrating the system's full capabilities and the value of integrating AI into the boot process. **END** 🐧

**By: Anisha Ghosh**

The author is a cybersecurity researcher and an active contributor to the open source communities and repositories. She is interested in developing novel, scalable and secure systems.